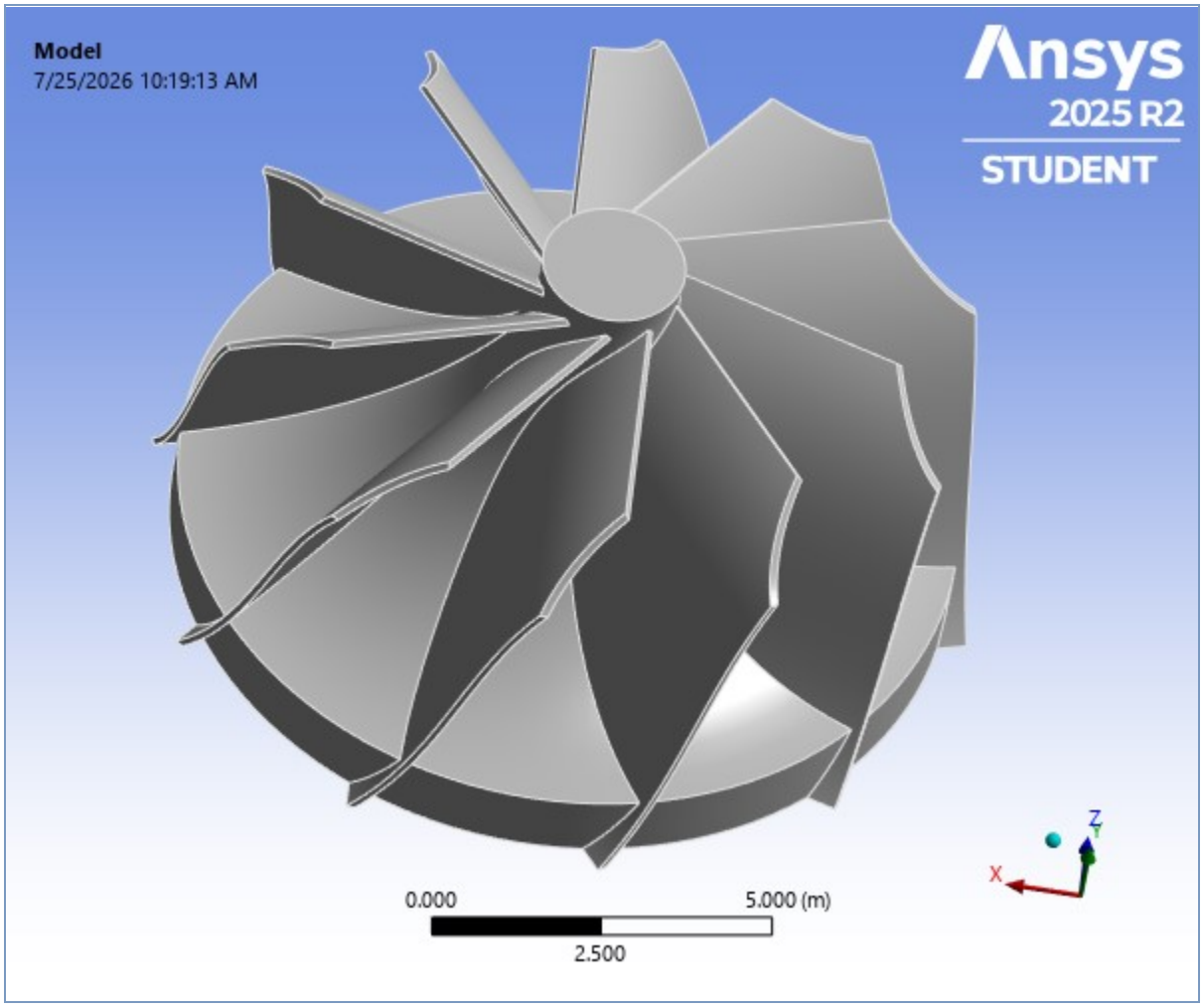
Project*

First Saved	Saturday, July 25, 2026
Last Saved	Saturday, July 25, 2026
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
- **Model (A4, B4)**
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    - Geometry Import (A3, B3)
  - Geometry
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  - Coordinate Systems
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    - Initial Conditions
      - Modal (None)
    - Analysis Settings
    - Loads
    - Imported Load (A6)
      - Imported Body Temperature
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      - Solution Information
      - Results
  - **Transient Thermal (A5)**
    - Initial Temperature
    - Analysis Settings
    - Convection
    - Solution (A6)
      - Solution Information
        - Result Charts
      - Temperature
- **Material Data**
  - Titanium Alloy

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4, B4)

TABLE 2
Model (A4, B4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3

Model (A4, B4) > Geometry Imports > Geometry Import (A3, B3)

Object Name	*Geometry Import (A3, B3)*
State	Solved
Definition	
Source	C:\Users\User\AppData\Local\Temp\WB_User_6476_2\wbnew_files\dp0\SYS-2\DM\SYS-2.scdocx
Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4, B4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	C:\Users\User\AppData\Local\Temp\WB_User_6476_2\wbnew_files\dp0\SYS-2\DM\SYS-2.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	12.256 m
Length Y	6.5023 m

Length Z	12.135 m
Properties	
Volume	224.71 m³
Mass	1.0022e+006 kg
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	14943
Elements	7690
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4, B4) > Geometry > Parts

Object Name	*SYS-2\Solid*
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System

Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Titanium Alloy
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	12.256 m
Length Y	6.5023 m
Length Z	12.135 m
Properties	
Volume	224.71 m³
Mass	1.0022e+006 kg
Centroid X	-5.2206e-005 m
Centroid Y	1.5867 m
Centroid Z	-5.6754e-006 m
Moment of Inertia Ip1	8.305e+006 kg·m²
Moment of Inertia Ip2	1.2251e+007 kg·m²
Moment of Inertia Ip3	8.3051e+006 kg·m²
Statistics	
Nodes	14943
Elements	7690
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:175.143.175	

TABLE 6
Model (A4, B4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4, B4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]

Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Mesh

TABLE 8
Model (A4, B4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	18.432 m
Average Surface Area	11.118 m²
Minimum Edge Length	0.11365 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No

Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	14943
Elements	7690
Show Detailed Statistics	No

Transient (B5)

TABLE 9
Model (A4, B4) > Analysis

Object Name	*Transient (B5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Transient
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 10
Model (A4, B4) > Transient (B5) > Initial Conditions

Object Name	*Initial Conditions*
State	Fully Defined

TABLE 11
Model (A4, B4) > Transient (B5) > Initial Conditions > Initial Condition

Object Name	*Modal (None)*
State	Fully Defined
Definition	
Modal Environment	None Available
Pre-Stress Environment	None

TABLE 12
Model (A4, B4) > Transient (B5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	5.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Time
Initial Time Step	1. s
Minimum Time Step	0.1 s
Maximum Time Step	1. s
Time Integration	On
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off

Large Deflection	Off
App. Based Settings	Moderate Speed Dynamics
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
--Energy Dissipation Ratio	1.e-004
Advanced	
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Damping Controls	
Stiffness Coefficient Define By	Direct Input
Stiffness Coefficient	0.
Mass Coefficient	0.
Analysis Data Management	
Solver Files Directory	C:\Users\User\AppData\Local\Temp\WB_User_6476_2\wbnew_files\dp0\SYS-3\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 13
Model (A4, B4) > Transient (B5) > Analysis Settings
Step-Specific "Step Controls"

Step	Step End Time	Initial Time Step	Carry Over Time Step

1	1. s	1. s	
2	2. s	0.5 s	Off
3	3. s		
4	4. s		
5	5. s		

TABLE 14
Model (A4, B4) > Transient (B5) > Loads

Object Name	*Thermal Condition*	*Fixed Support*	*Pressure*
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	1 Body	1 Face	30 Faces
Definition			
Type	Thermal Condition	Fixed Support	Pressure
Magnitude	20. °C (step applied)		Tabular Data
Suppressed	No		
Define By			Normal To
Applied By			Surface Effect
Loaded Area			Deformed
Tabular Data			
Independent Variable			Time

FIGURE 1
Model (A4, B4) > Transient (B5) > Thermal Condition

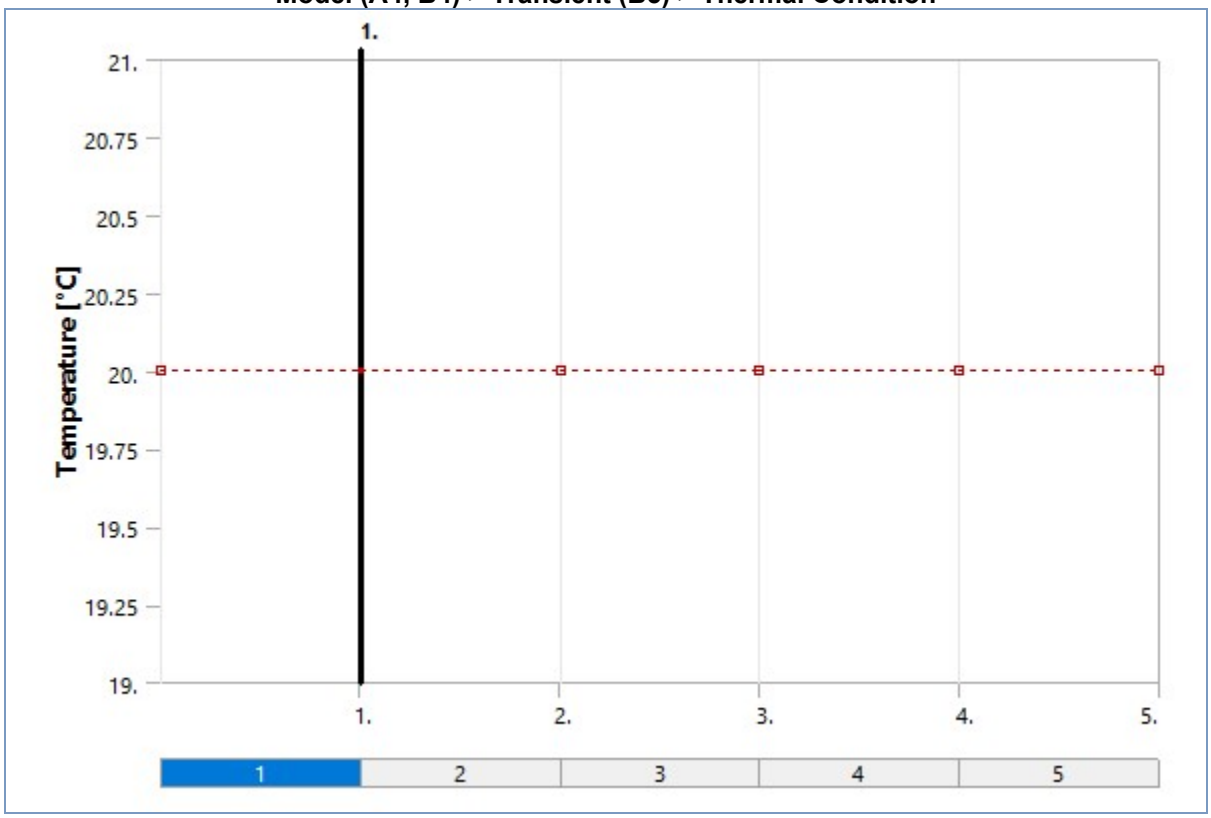


FIGURE 2
Model (A4, B4) > Transient (B5) > Pressure

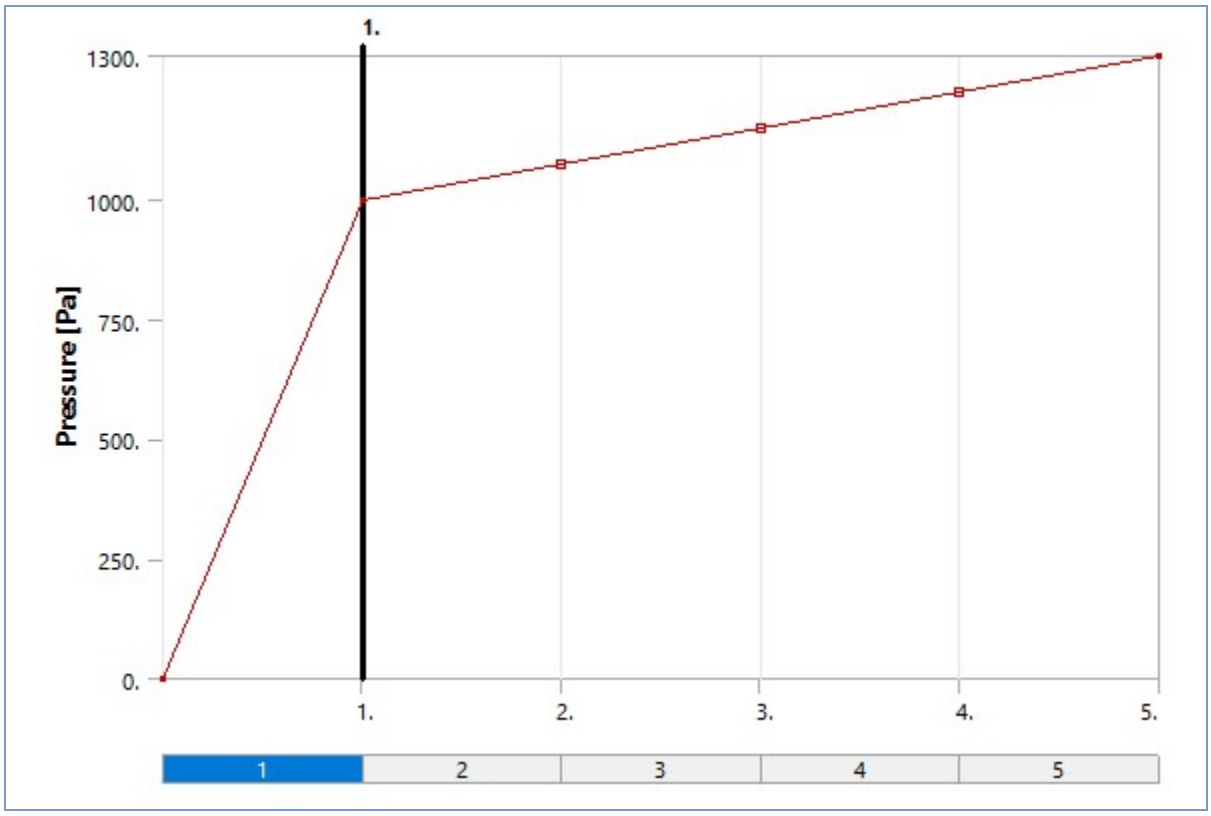


TABLE 15
Model (A4, B4) > Transient (B5) > Pressure

Steps	Time [s]	Pressure [Pa]
1	0.	0.
	1.	1000.
2	2.	= 1075.
3	3.	= 1150.
4	4.	= 1225.
5	5.	1300.

TABLE 16
Model (A4, B4) > Transient (B5) > Imported Load (A6)

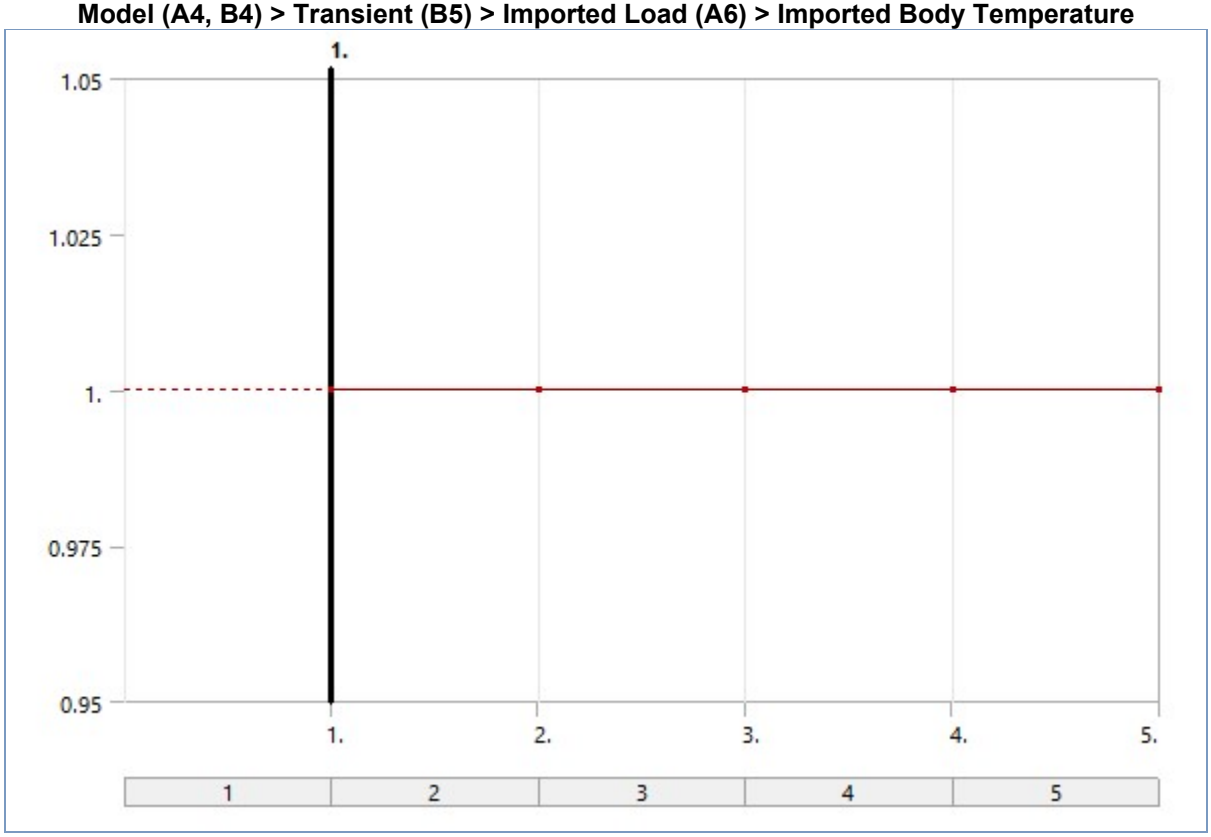
Object Name	*Imported Load (A6)*
State	Fully Defined
Definition	
Type	Imported Data
Interpolation Type	Mechanical Results Transfer
Suppressed	No
Source	A6::Solution 2
Data Management	
Delete Mapped Data Files	Yes

TABLE 17
Model (A4, B4) > Transient (B5) > Imported Load (A6) > Imported Body Temperature

Object Name	*Imported Body Temperature*
State	Solved
Scope	
Scoping Method	Geometry Selection

Geometry	1 Body
Definition	
Type	Imported Body Temperature
Tabular Loading	Program Controlled
Mapped Data	To Input File
Suppressed	No
Source Environment	Transient Thermal (A5)
Source Time	Worksheet

FIGURE 3
Model (A4, B4) > Transient (B5) > Imported Load (A6) > Imported Body Temperature



Model (A4, B4) > Transient (B5) > Imported Load (A6) > Imported Body Temperature

	Source Time (s)	Analysis Time (s)	Scale	Offset (Δ°C)
1	End Time	1	1	0
*				

Solution (B6)

TABLE 18
Model (A4, B4) > Transient (B5) > Solution

Object Name	*Solution (B6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	2 m 20 s

MAPDL Memory Used	169. MB
MAPDL Result File Size	115.94 MB
Post Processing	
Beam Section Results	No

TABLE 19
Model (A4, B4) > Transient (B5) > Solution (B6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 20
Model (A4, B4) > Transient (B5) > Solution (B6) > Results

Object Name	*Equivalent Elastic Strain*	*Equivalent Stress*	*Total Deformation*	*Total Acceleration*
State	Solved			
Scope				
Scoping Method	Geometry Selection			
Geometry	All Bodies			
Definition				
Type	Equivalent Elastic Strain	Equivalent (von-Mises) Stress	Total Deformation	Total Acceleration
By	Time			
Display Time	Last			
Separate Data by Entity	No			
Calculate Time History	Yes			
Identifier				
Suppressed	No			
Integration Point Results				
Display Option	Averaged			
Average Across Bodies	No			
Results				
Minimum	1.3816e-006 m/m	64554 Pa	0. m	0. m/s²
Maximum	1.318e-004 m/m	1.4218e+007 Pa	7.8567e-004 m	3.2796e-004 m/s²
Average	2.407e-005 m/m	1.5671e+006 Pa	1.4025e-004 m	1.2028e-004 m/s²
Minimum Occurs On	SYS-2\Solid			

Maximum Occurs On	SYS-2\Solid			
Minimum Value Over Time				
Minimum	1.3708e-006 m/m	61314 Pa	0. m	0. m/s²
Maximum	1.3816e-006 m/m	64554 Pa	0. m	0. m/s²
Maximum Value Over Time				
Minimum	1.318e-004 m/m	1.4218e+007 Pa	7.2323e-004 m	1.7278e-004 m/s²
Maximum	1.3187e-004 m/m	1.4225e+007 Pa	7.8567e-004 m	2.4868e-002 m/s²
Information				
Time	5. s			
Load Step	5			
Substep	6			
Iteration Number	25			

FIGURE 4
Model (A4, B4) > Transient (B5) > Solution (B6) > Equivalent Elastic Strain

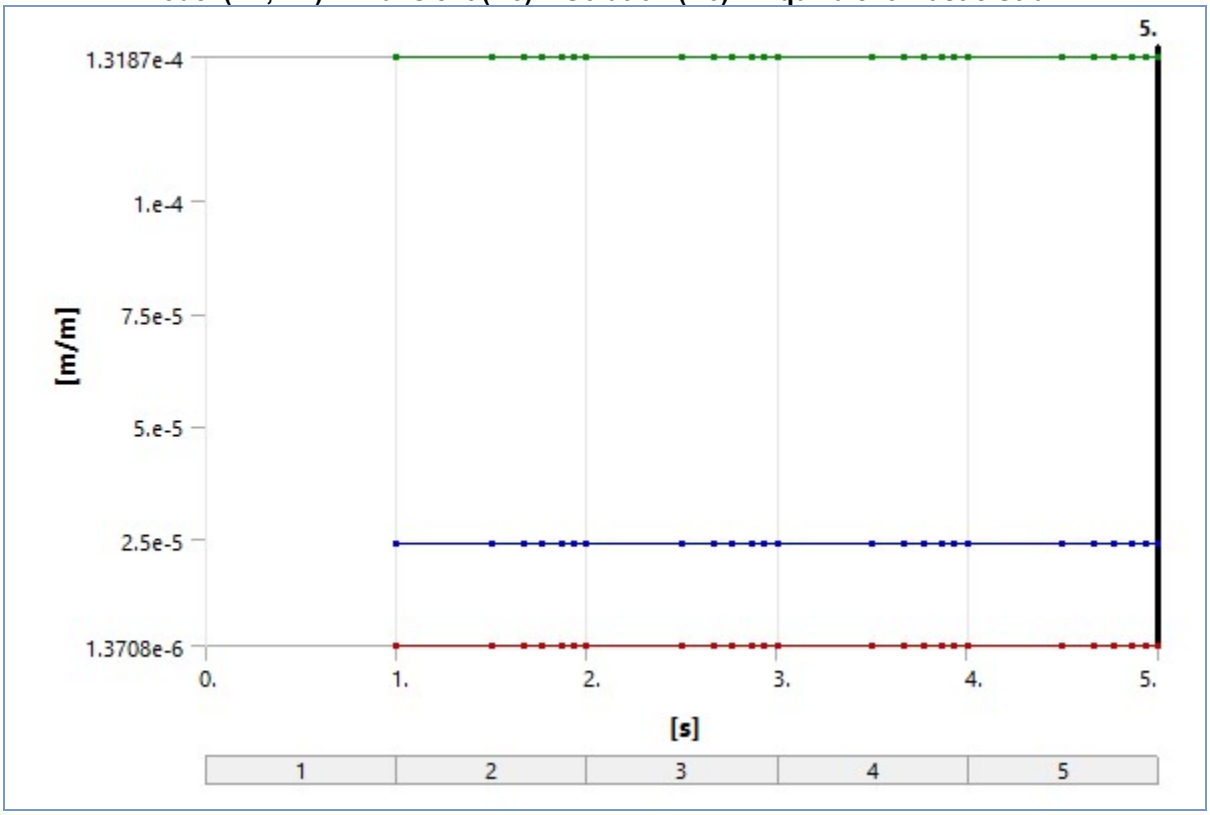


TABLE 21
Model (A4, B4) > Transient (B5) > Solution (B6) > Equivalent Elastic Strain

Time [s]	Minimum [m/m]	Maximum [m/m]	Average [m/m]
1.	1.3708e-006	1.3187e-004	2.3996e-005
1.5	1.3721e-006	1.3186e-004	2.4005e-005
1.6667	1.3722e-006		
1.7667	1.3732e-006	1.3185e-004	2.4012e-005
1.8667	1.3724e-006	1.3186e-004	2.4007e-005
1.9333	1.3738e-006	1.3185e-004	2.4016e-005
2.	1.3726e-006		2.4009e-005
2.5	1.3751e-006		2.4025e-005
2.6667	1.3747e-006		2.4023e-005

2.7667	1.3755e-006	1.3184e-004	2.4027e-005
2.8667	1.3754e-006		2.4029e-005
2.9333	1.3757e-006		
3.	1.3759e-006	1.3183e-004	2.4032e-005
3.5	1.3772e-006		2.4039e-005
3.6667	1.3777e-006	1.3182e-004	2.4044e-005
3.7667	1.378e-006		2.4045e-005
3.8667	1.3783e-006		2.4048e-005
3.9333	1.3785e-006		2.4049e-005
4.	1.3786e-006		2.405e-005
4.5	1.3801e-006	1.3181e-004	2.406e-005
4.6667	1.3806e-006		2.4063e-005
4.7667	1.3809e-006	1.318e-004	2.4066e-005
4.8667	1.3812e-006		2.4067e-005
4.9333	1.3814e-006		2.4069e-005
5.	1.3816e-006		2.407e-005

FIGURE 5
Model (A4, B4) > Transient (B5) > Solution (B6) > Equivalent Elastic Strain > Figure

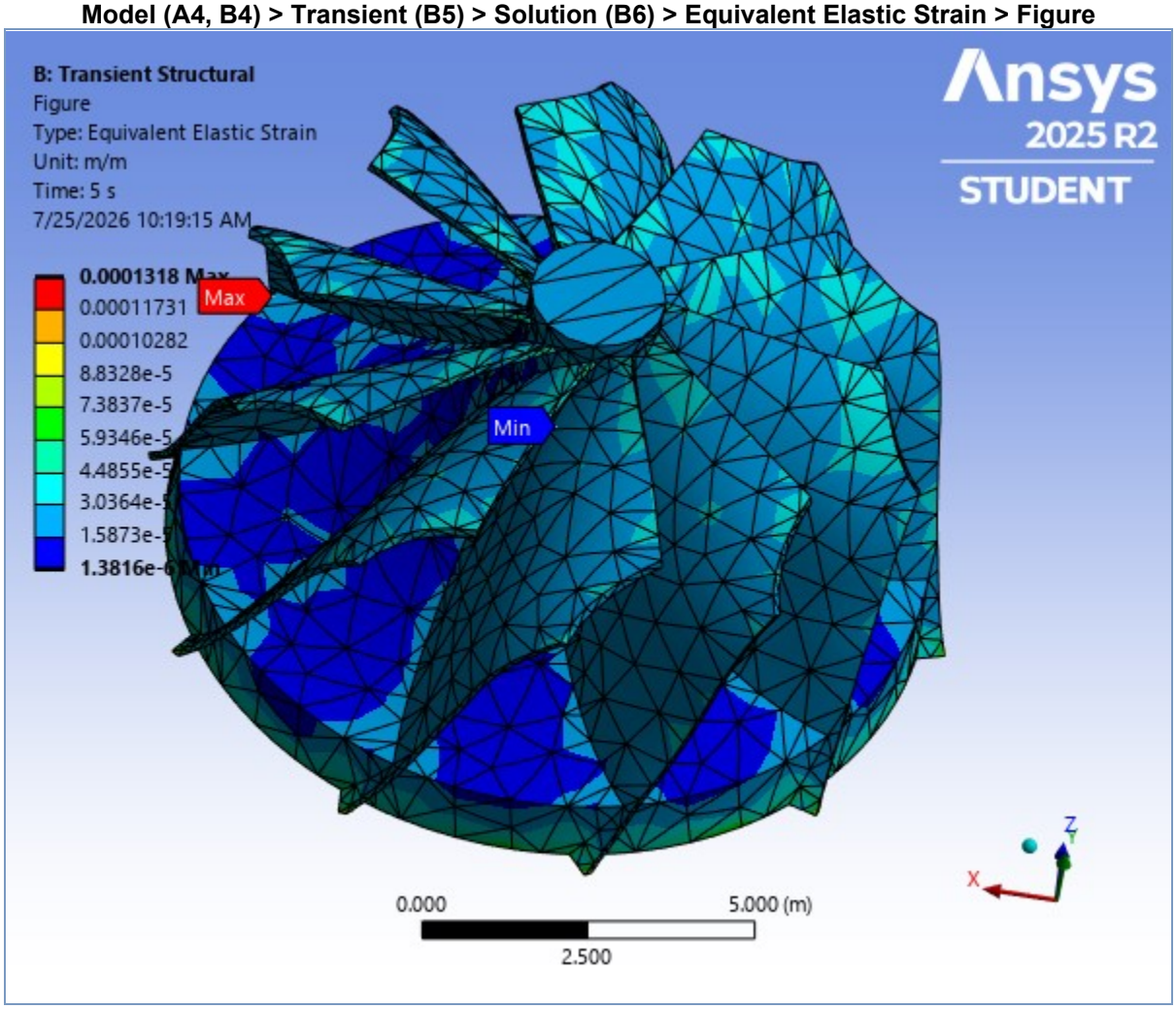


FIGURE 6
Model (A4, B4) > Transient (B5) > Solution (B6) > Equivalent Stress

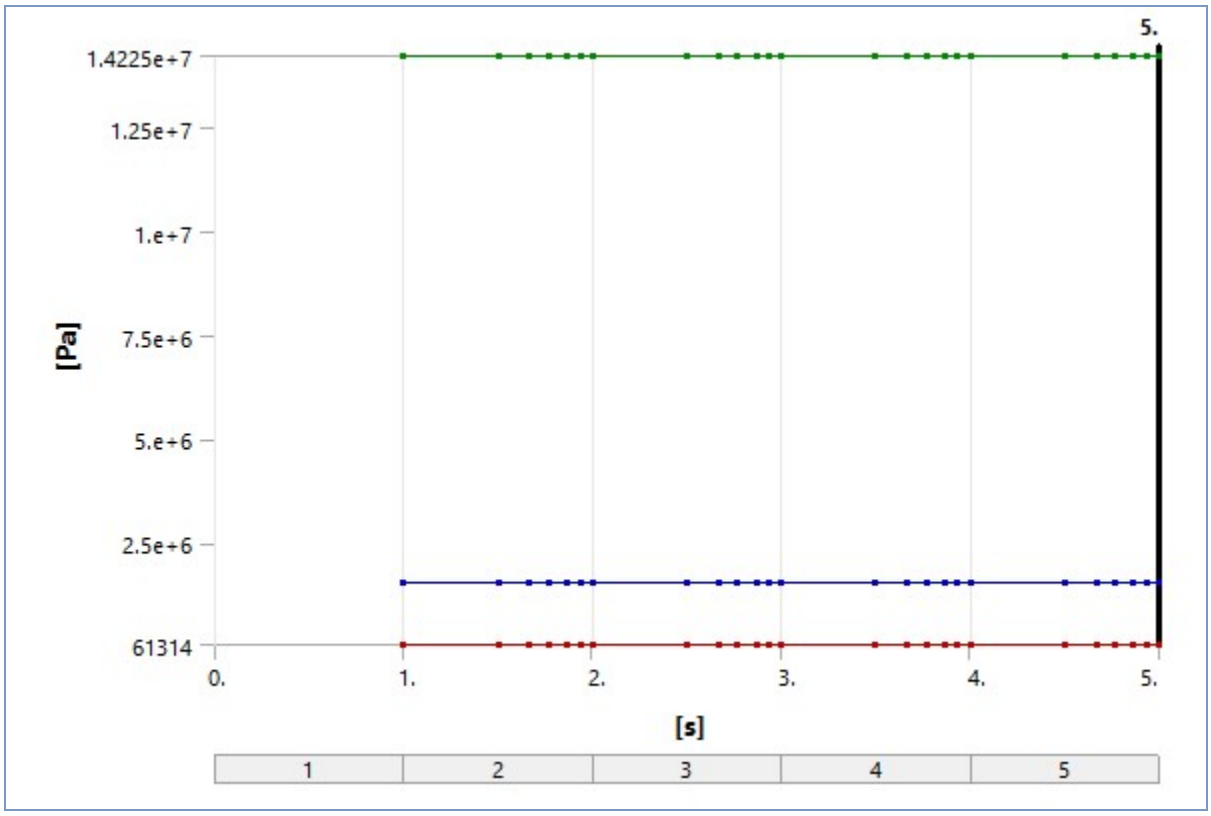


TABLE 22
Model (A4, B4) > Transient (B5) > Solution (B6) > Equivalent Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	61314	1.4225e+007	1.554e+006
1.5	61727	1.4224e+007	1.5556e+006
1.6667	61769		1.5558e+006
1.7667	62043		1.5568e+006
1.8667	61830		1.5561e+006
1.9333	62241	1.4223e+007	1.5575e+006
2.	61906	1.4224e+007	1.5565e+006
2.5	62633	1.4222e+007	1.5591e+006
2.6667	62538	1.4223e+007	1.559e+006
2.7667	62759	1.4222e+007	1.5596e+006
2.8667	62770		1.5599e+006
2.9333	62838		
3.	62912		1.5605e+006
3.5	63291	1.4221e+007	1.5618e+006
3.6667	63449		1.5626e+006
3.7667	63519	1.422e+007	1.5628e+006
3.8667	63604		1.5632e+006
3.9333	63666		1.5634e+006
4.	63706		1.5636e+006
4.5	64139	1.4219e+007	1.5654e+006
4.6667	64266		1.5659e+006
4.7667	64359		1.5663e+006
4.8667	64437	1.4218e+007	1.5666e+006
4.9333	64496		1.5669e+006
5.	64554		1.5671e+006

FIGURE 7
Model (A4, B4) > Transient (B5) > Solution (B6) > Equivalent Stress > Figure

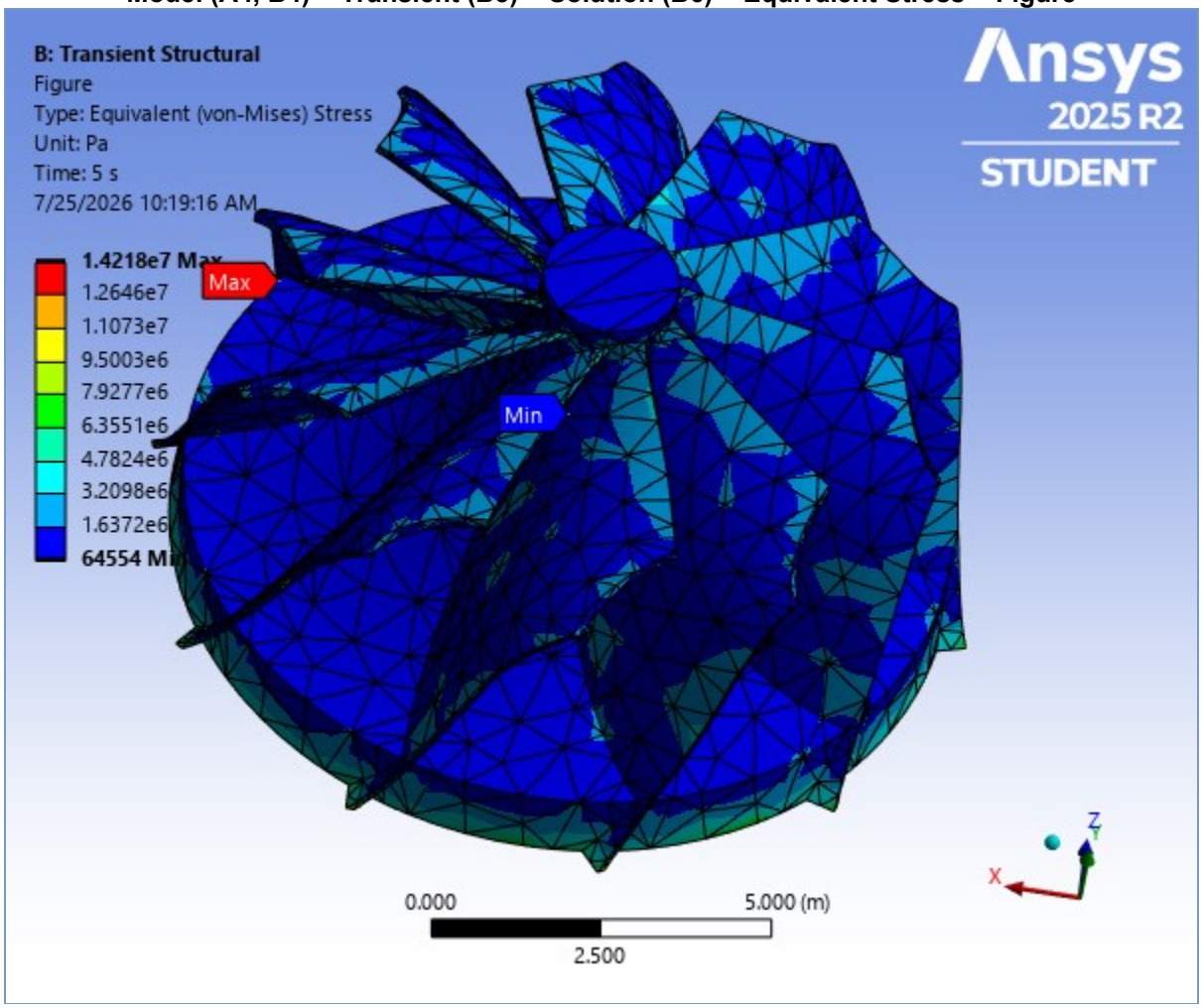


FIGURE 8
Model (A4, B4) > Transient (B5) > Solution (B6) > Total Deformation

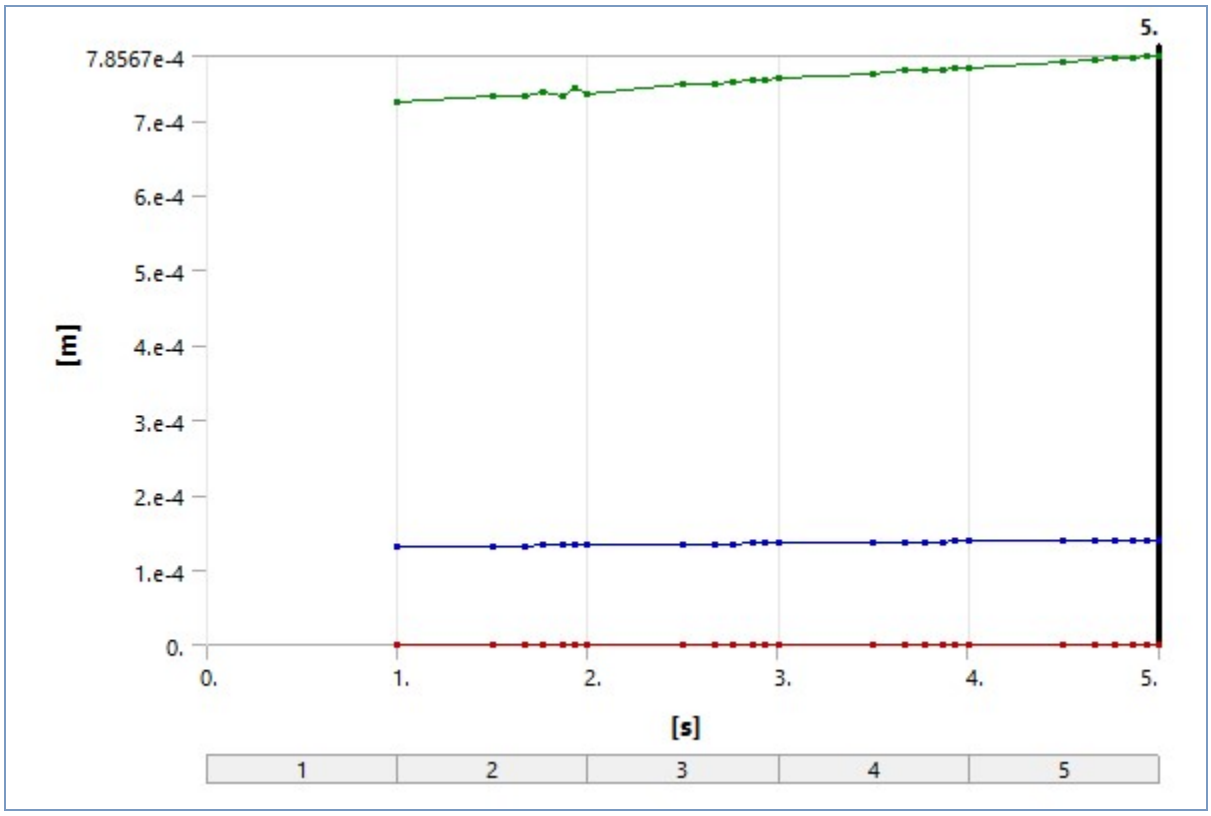


TABLE 23
Model (A4, B4) > Transient (B5) > Solution (B6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.		7.2323e-004	1.3075e-004
1.5		7.3165e-004	1.3197e-004
1.6667		7.3196e-004	1.3214e-004
1.7667		7.3855e-004	1.3286e-004
1.8667		7.3271e-004	1.3237e-004
1.9333		7.4251e-004	1.3338e-004
2.		7.3473e-004	1.327e-004
2.5		7.4954e-004	1.3453e-004
2.6667		7.4757e-004	1.345e-004
2.7667		7.5135e-004	1.3492e-004
2.8667		7.5271e-004	1.3517e-004
2.9333		7.5228e-004	1.3515e-004
3.	0.	7.5593e-004	1.356e-004
3.5		7.6116e-004	1.3652e-004
3.6667		7.6564e-004	1.3711e-004
3.7667		7.6594e-004	1.3722e-004
3.8667		7.6823e-004	1.3753e-004
3.9333		7.6905e-004	1.3768e-004
4.		7.6988e-004	1.3781e-004
4.5		7.7806e-004	1.3907e-004
4.6667		7.8036e-004	1.3943e-004
4.7667		7.8221e-004	1.3971e-004
4.8667		7.8351e-004	1.3992e-004
4.9333		7.8478e-004	1.401e-004
5.		7.8567e-004	1.4025e-004

FIGURE 9
Model (A4, B4) > Transient (B5) > Solution (B6) > Total Deformation > Figure

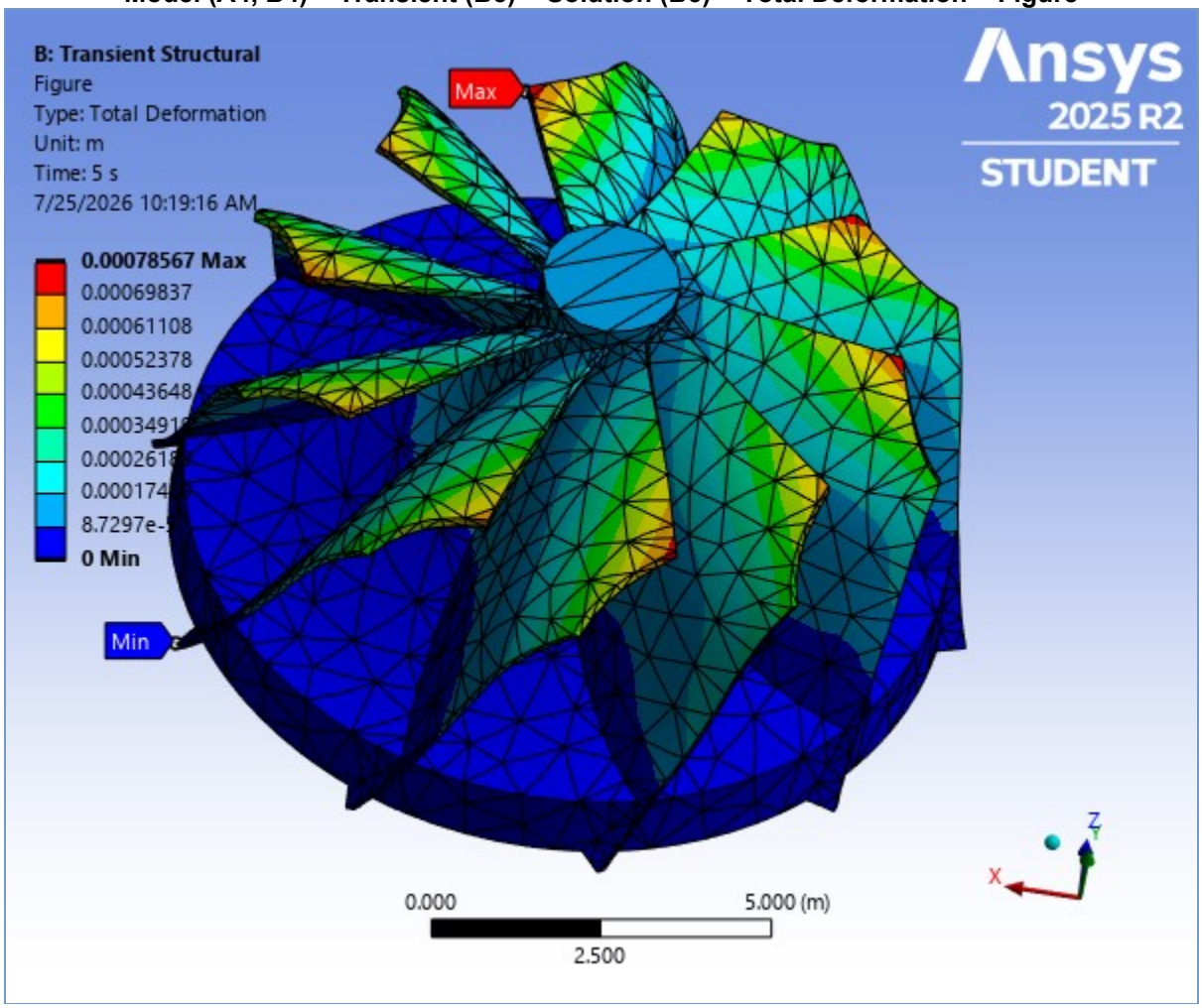


FIGURE 10
Model (A4, B4) > Transient (B5) > Solution (B6) > Total Acceleration

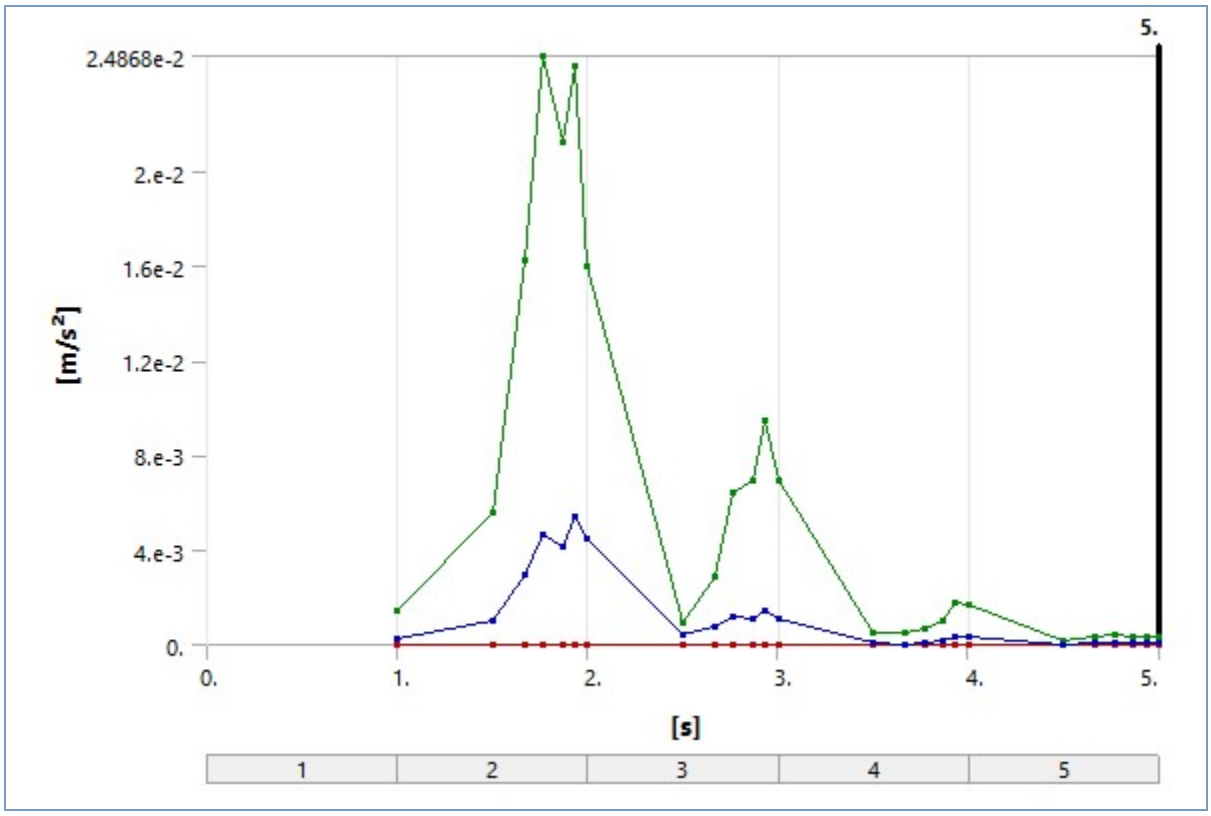
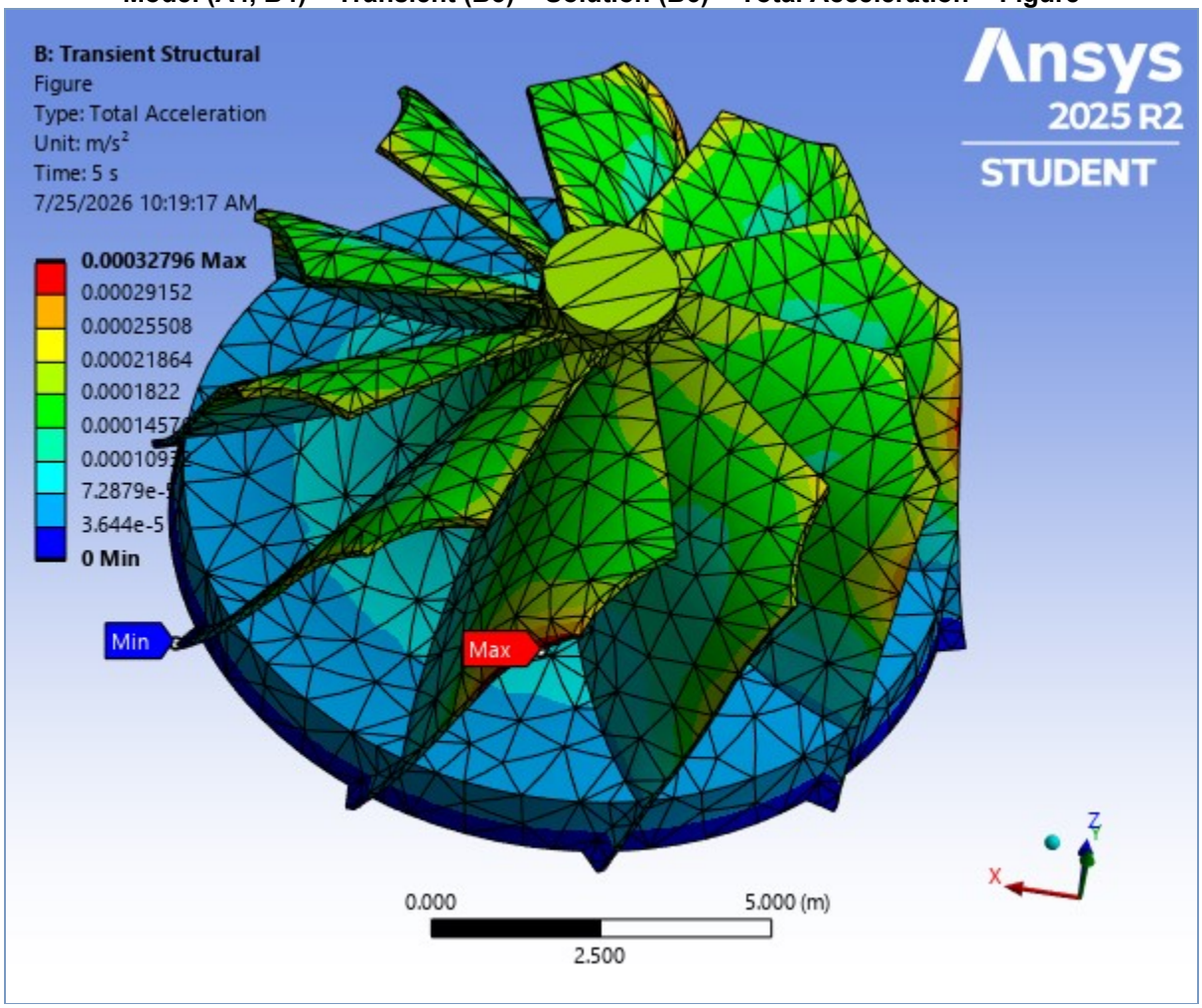


TABLE 24
Model (A4, B4) > Transient (B5) > Solution (B6) > Total Acceleration

Time [s]	Minimum [m/s²]	Maximum [m/s²]	Average [m/s²]
1.	0.	1.4345e-003	2.5934e-004
1.5		5.6042e-003	1.0156e-003
1.6667		1.6207e-002	2.9523e-003
1.7667		2.4868e-002	4.6393e-003
1.8667		2.122e-002	4.1845e-003
1.9333		2.4413e-002	5.3792e-003
2.		1.5947e-002	4.4783e-003
2.5		9.1227e-004	3.9864e-004
2.6667		2.8875e-003	7.7643e-004
2.7667		6.4008e-003	1.2082e-003
2.8667		6.9679e-003	1.1329e-003
2.9333		9.4821e-003	1.4717e-003
3.		6.9397e-003	1.138e-003
3.5		4.7239e-004	7.4486e-005
3.6667		5.3374e-004	3.8781e-005
3.7667		6.8864e-004	1.2244e-004
3.8667		1.0283e-003	1.8866e-004
3.9333		1.7973e-003	3.2818e-004
4.		1.7008e-003	3.1497e-004
4.5		1.7278e-004	3.405e-005
4.6667		3.7534e-004	8.0233e-005
4.7667		4.5281e-004	1.1272e-004
4.8667		2.9723e-004	9.5113e-005
4.9333		3.255e-004	1.2494e-004
5.		3.2796e-004	1.2028e-004

FIGURE 11
Model (A4, B4) > Transient (B5) > Solution (B6) > Total Acceleration > Figure



Transient Thermal (A5)

TABLE 25
Model (A4, B4) > Analysis

Object Name	*Transient Thermal (A5)*
State	Solved
Definition	
Physics Type	Thermal
Analysis Type	Transient
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 26
Model (A4, B4) > Transient Thermal (A5) > Initial Condition

Object Name	*Initial Temperature*
State	Fully Defined
Definition	
Initial Temperature	Uniform Temperature
Initial Temperature Value	20. °C

TABLE 27
Model (A4, B4) > Transient Thermal (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	5.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Initial Time Step	1.e-002 s
Minimum Time Step	1.e-003 s
Maximum Time Step	0.1 s
Time Integration	On
Solver Controls	
Solver Type	Program Controlled
Radiosity Controls	
Radiosity Solver	Program Controlled
Flux Convergence	1.e-004
Maximum Iteration	1000.
Solver Tolerance	0.1 W/m²
Over Relaxation	0.1
Hemicube Resolution	10.
Nonlinear Controls	
Heat Convergence	Program Controlled
Temperature Convergence	Program Controlled
Line Search	Program Controlled
Nonlinear Formulation	Program Controlled
Advanced	
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Calculate Thermal Flux	Yes
Contact Data	Yes
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\User\AppData\Local\Temp\WB_User_6476_2\wbnew_files\dp0\SYS-2\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 28
Model (A4, B4) > Transient Thermal (A5) > Analysis Settings
Step-Specific "Step Controls"

Step	Step End Time
1	1. s
2	2. s
3	3. s
4	4. s
5	5. s

TABLE 29
Model (A4, B4) > Transient Thermal (A5) > Loads

Object Name	*Convection*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	32 Faces
Definition	
Type	Convection
Film Coefficient	1550. W/m²·°C (step applied)
Ambient Temperature	40. °C (step applied)
Convection Matrix	Program Controlled
Element APDL Name	
Suppressed	No

FIGURE 12
Model (A4, B4) > Transient Thermal (A5) > Convection

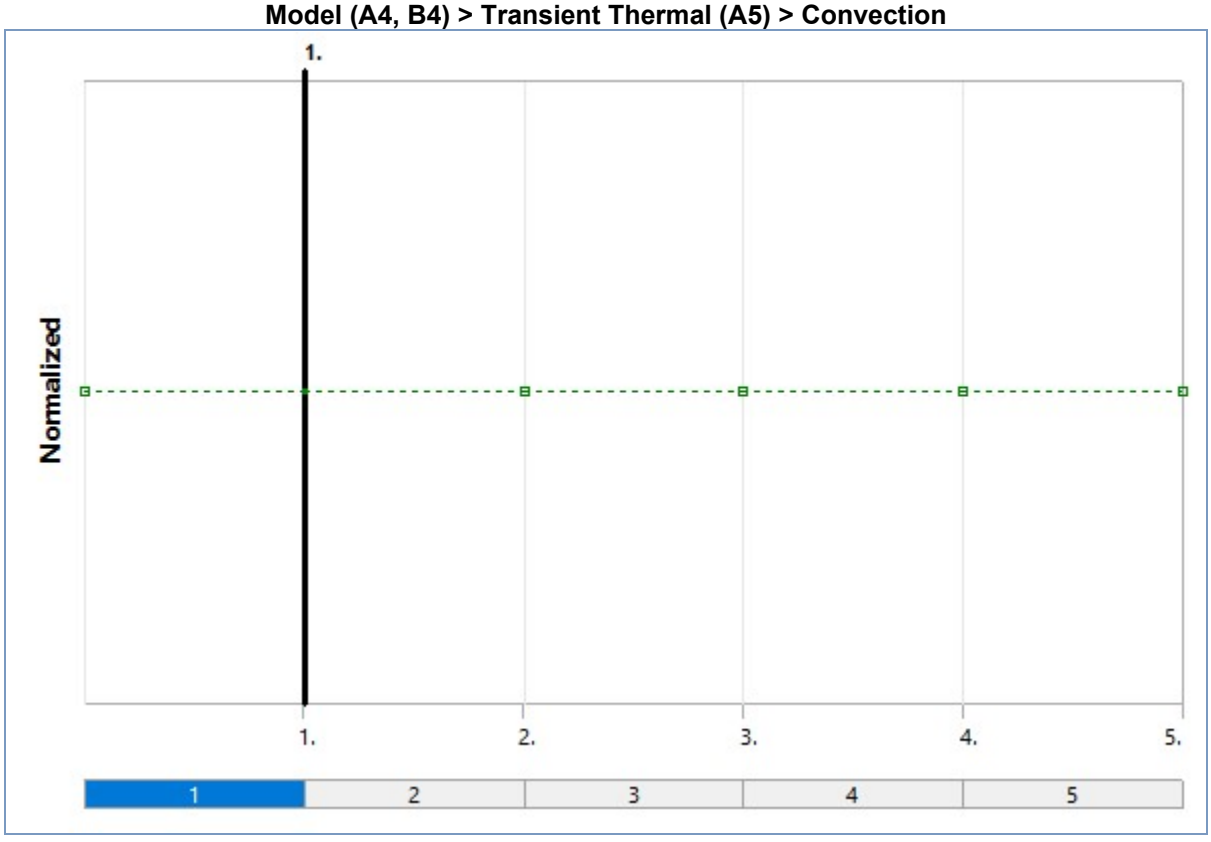


TABLE 30
Model (A4, B4) > Transient Thermal (A5) > Convection

Steps	Time [s]	Convection Coefficient [W/m²·°C]	Temperature [°C]
1	0.	= 1550.	= 40.
	1.	1550.	40.
2	2.	= 1550.	= 40.
3	3.		
4	4.		
5	5.		

Solution (A6)

TABLE 31
Model (A4, B4) > Transient Thermal (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	48. s
MAPDL Memory Used	204. MB
MAPDL Result File Size	110.62 MB
Post Processing	
Beam Section Results	No

TABLE 32
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 33
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Result Charts

Object Name	*Temperature - Global Maximum*	*Temperature - Global Minimum*
State	Solved	
Scope		
Scoping Method	Global Maximum	Global Minimum
Definition		
Type	Temperature	
Suppressed	No	
Results		
Minimum	64.659 °C	14.366 °C

Maximum	71.747 °C	27.35 °C

FIGURE 13
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Maximum

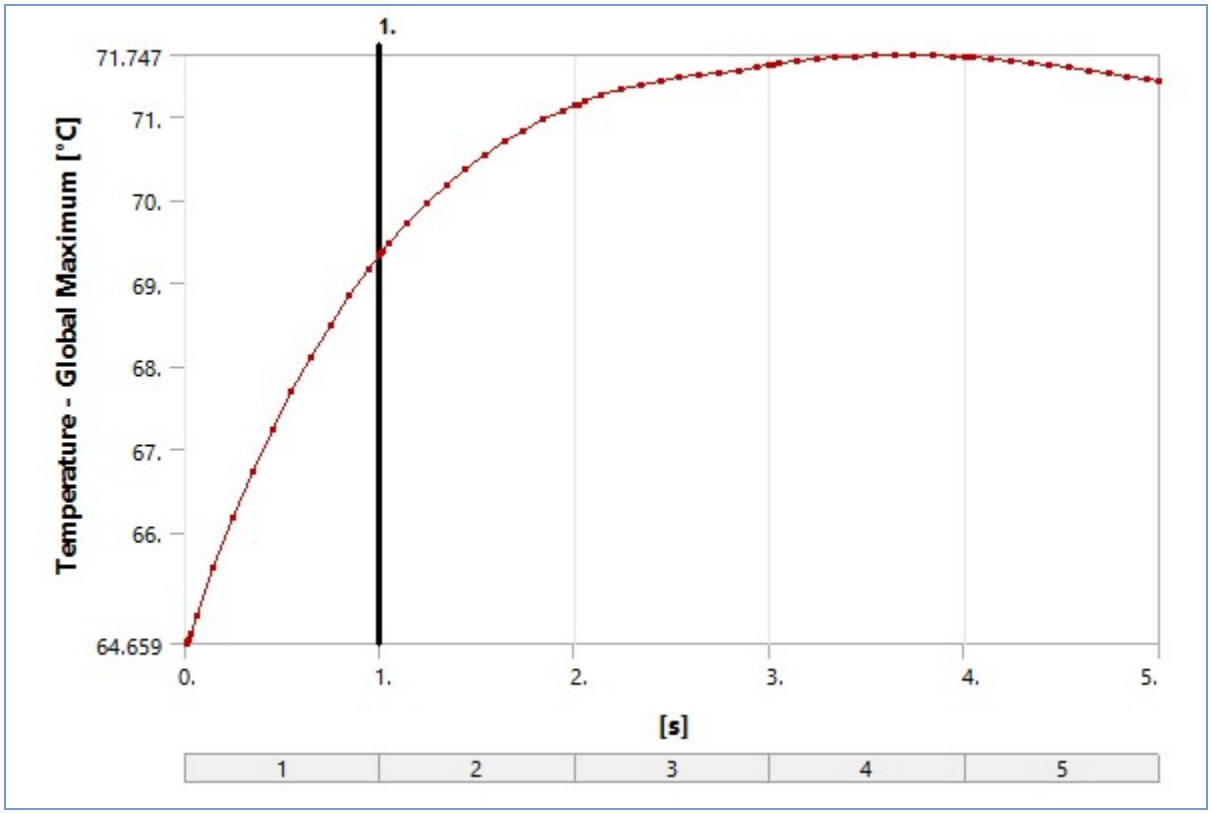


FIGURE 14
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Minimum

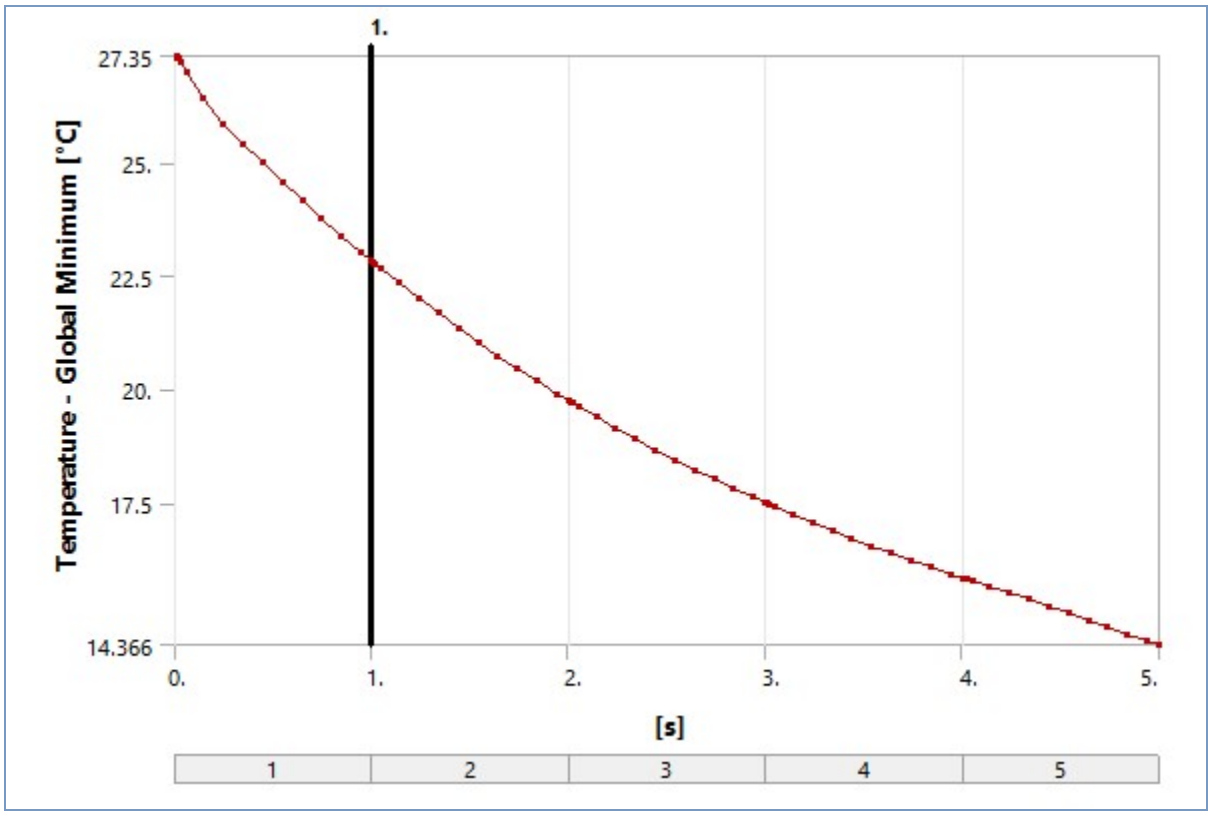


TABLE 34
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Results

Object Name	*Temperature*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Temperature
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Results	
Minimum	14.366 °C
Maximum	71.423 °C
Average	31.922 °C
Minimum Occurs On	SYS-2\Solid
Maximum Occurs On	SYS-2\Solid
Minimum Value Over Time	
Minimum	14.366 °C
Maximum	27.35 °C
Maximum Value Over Time	
Minimum	64.659 °C
Maximum	71.747 °C
Information	

Time	5. s
Load Step	5
Substep	13
Iteration Number	67

FIGURE 15
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Temperature

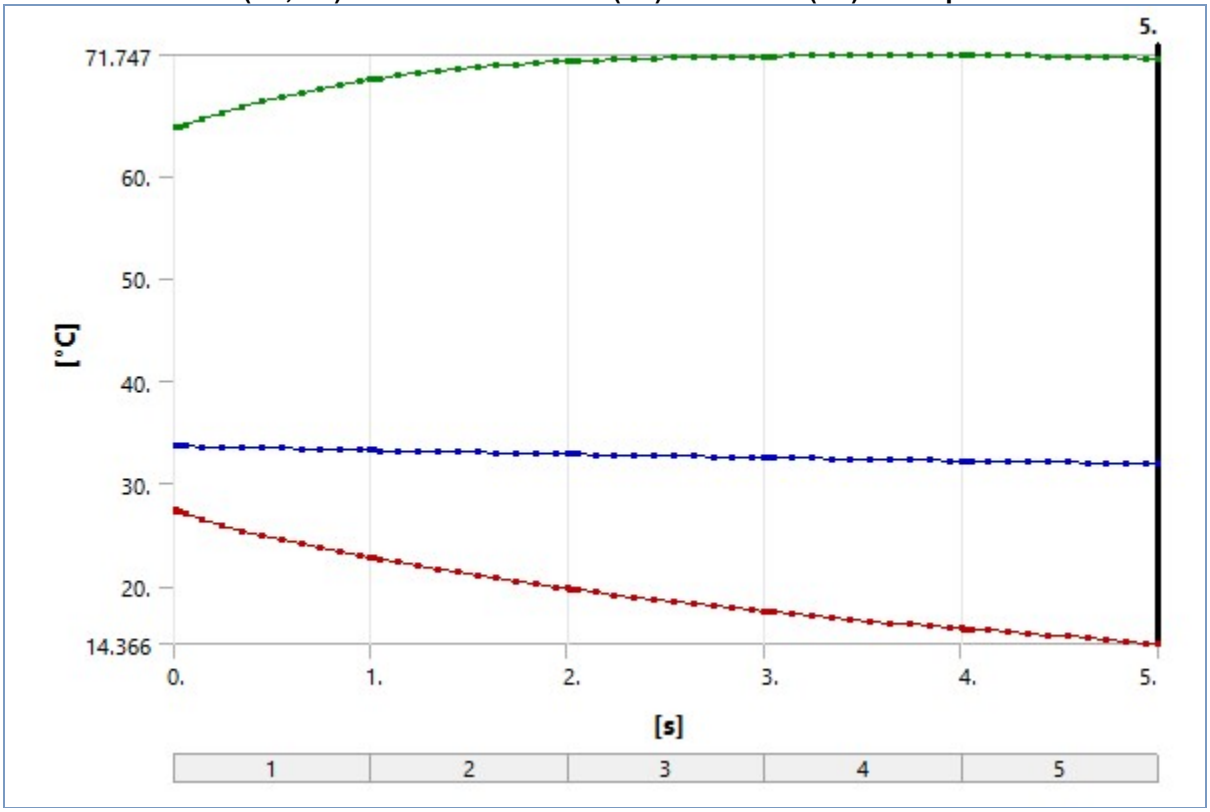
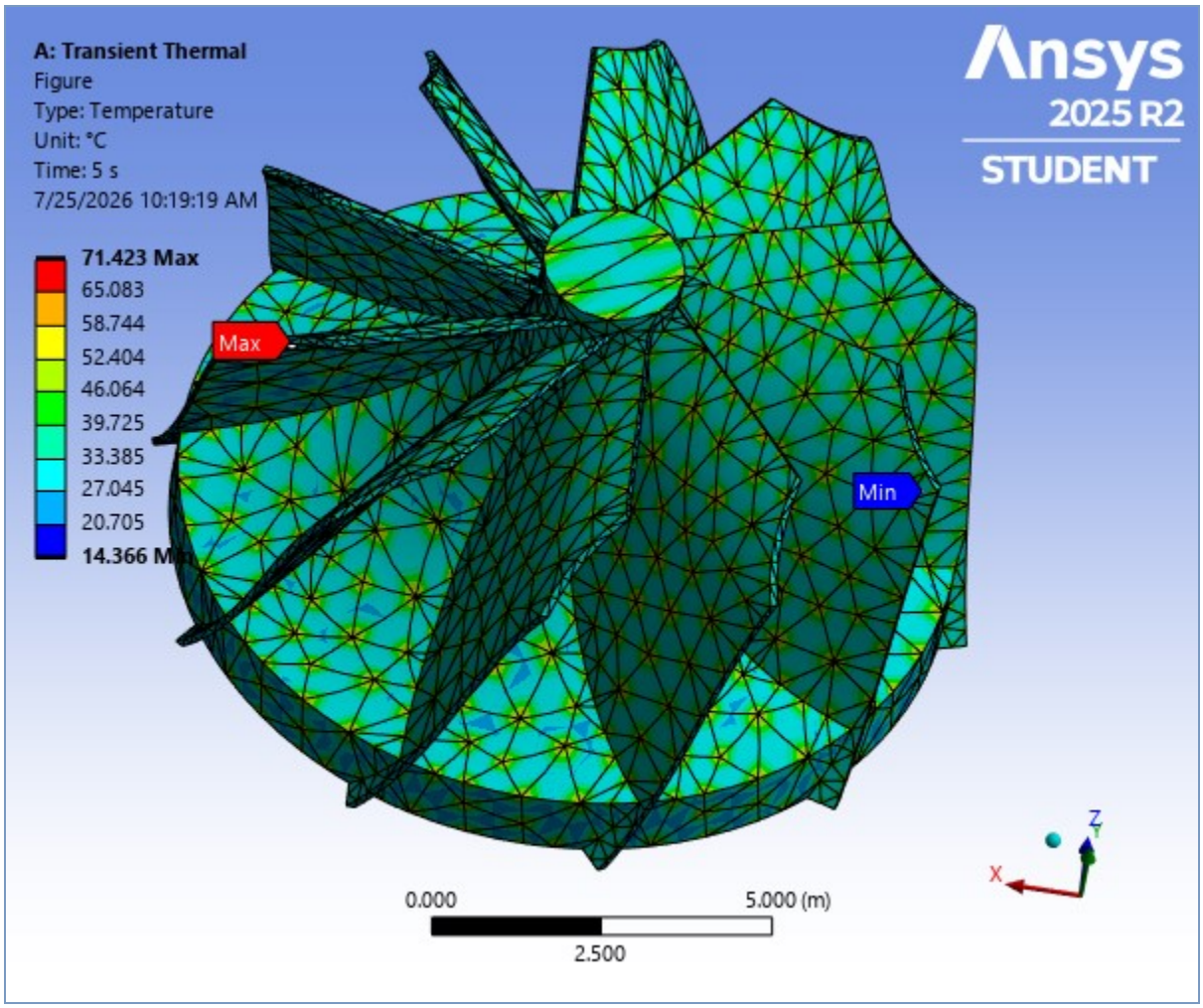


TABLE 35
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Temperature

Time [s]	Minimum [°C]	Maximum [°C]	Average [°C]
1.e-002	27.35	64.659	33.647
1.3333e-002	27.323	64.684	33.646
1.6667e-002	27.297	64.708	33.644
2.6667e-002	27.218	64.781	33.64
5.6667e-002	26.995	64.992	33.626
0.14667	26.414	65.58	33.585
0.24667	25.863	66.178	33.54
0.34667	25.393	66.726	33.495
0.44667	24.99	67.228	33.451
0.54667	24.57	67.688	33.408
0.64667	24.15	68.108	33.366
0.74667	23.752	68.492	33.324
0.84667	23.375	68.843	33.282
0.94667	23.016	69.163	33.242
1.	22.83	69.325	33.22
1.01	22.795	69.355	33.216
1.02	22.761	69.385	33.212
1.05	22.658	69.471	33.2

1.14	22.359	69.71	33.164
1.24	22.013	69.951	33.124
1.34	21.679	70.168	33.086
1.44	21.357	70.364	33.047
1.54	21.046	70.541	33.009
1.64	20.746	70.699	32.972
1.74	20.456	70.839	32.935
1.84	20.175	70.964	32.898
1.94	19.904	71.073	32.862
2.	19.744	71.134	32.84
2.01	19.718	71.144	32.837
2.02	19.691	71.153	32.833
2.05	19.613	71.181	32.823
2.14	19.384	71.255	32.791
2.24	19.138	71.325	32.756
2.34	18.9	71.384	32.721
2.44	18.669	71.432	32.687
2.54	18.446	71.47	32.653
2.64	18.229	71.499	32.62
2.74	18.019	71.519	32.587
2.84	17.815	71.552	32.554
2.94	17.617	71.598	32.522
3.	17.5	71.623	32.502
3.01	17.481	71.627	32.499
3.02	17.462	71.631	32.496
3.05	17.404	71.643	32.486
3.14	17.236	71.672	32.458
3.24	17.055	71.698	32.427
3.34	16.88	71.718	32.395
3.44	16.709	71.733	32.365
3.54	16.543	71.742	32.334
3.64	16.381	71.747	32.304
3.74	16.224	71.746	32.274
3.84	16.071	71.741	32.245
3.94	15.923	71.732	32.215
4.	15.835	71.725	32.198
4.01	15.821	71.723	32.195
4.02	15.806	71.722	32.192
4.05	15.763	71.718	32.184
4.14	15.636	71.702	32.158
4.24	15.499	71.681	32.129
4.34	15.366	71.657	32.101
4.44	15.225	71.629	32.073
4.54	15.065	71.599	32.046
4.64	14.907	71.565	32.018
4.74	14.753	71.529	31.991
4.84	14.602	71.49	31.964
4.94	14.454	71.449	31.937
5.	14.366	71.423	31.922

FIGURE 16
Model (A4, B4) > Transient Thermal (A5) > Solution (A6) > Temperature > Figure



Material Data

Titanium Alloy

TABLE 36
Titanium Alloy > Constants

Density	4460 kg m^-3
Coefficient of Thermal Expansion	9.4e-006 C^-1
Specific Heat	522 J kg^-1 C^-1
Thermal Conductivity	7 W m^-1 C^-1
Resistivity	1.7e-006 kg m^3 A^-2 s^-3

TABLE 37
Titanium Alloy > Color

Red	Green	Blue
88	72	117

TABLE 38
Titanium Alloy > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
1.07e+009

TABLE 39
Titanium Alloy > Compressive Yield Strength

Compressive Yield Strength Pa
9.3e+008

TABLE 40
Titanium Alloy > Tensile Yield Strength

Tensile Yield Strength Pa
9.3e+008

TABLE 41
Titanium Alloy > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
1.07e+009

TABLE 42
Titanium Alloy > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 43
Titanium Alloy > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
1.1e+011	0.34	1.1458e+011	4.1045e+010	

TABLE 44
Titanium Alloy > Isotropic Relative Permeability

Relative Permeability
1